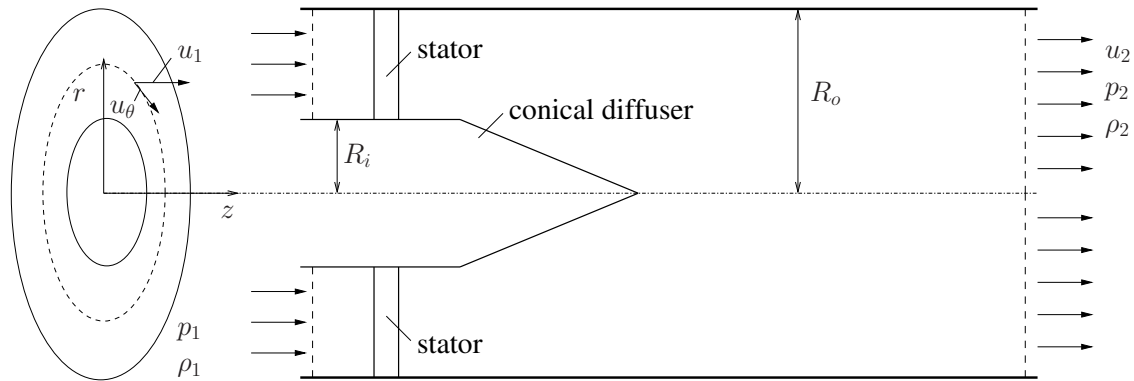


The diffuser module at the exit of a gas turbine consists of a **fixed** aerofoil cascade (stator), of inner and outer radii R_i and R_o , respectively, together with a conical diffuser, as depicted in the figure below. The gas enters the module with velocity field $\bar{v}_1 = u_1\bar{e}_z + u_\theta\bar{e}_\theta$, having both axial and azimuthal components, pressure p_1 and density ρ_1 , and leaves the module with a purely axial velocity field $\bar{v}_2 = u_2\bar{e}_z$, pressure p_2 and density ρ_2 .

For the analysis of the problem, one may introduce the following assumptions:

- The effect of gravity is entirely negligible.
- Flow properties at the entrance and exit sections are uniform.
- The air around the casing is at rest with uniform pressure p_a .
- The working gas behaves as a perfect gas (gas constant R_g , specific heat ratio γ).
- All the inner walls are adiabatic.

With the above simplifications, write the equations that determine the values of ρ_2 and u_2 , as well as the net force, $\bar{F}$, and torque, $\bar{M}$, acting on the diffuser module, in terms of the known quantities (p_a , p_1 , ρ_1 , p_2 , u_1 , u_θ , R_i , R_o , γ and R_g).



Integral conservation equations for an arbitrary control volume V_c :

$$\frac{d}{dt} \left[\int_{V_c(t)} \rho dV \right] + \int_{\Sigma_c(t)} \rho(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = 0$$

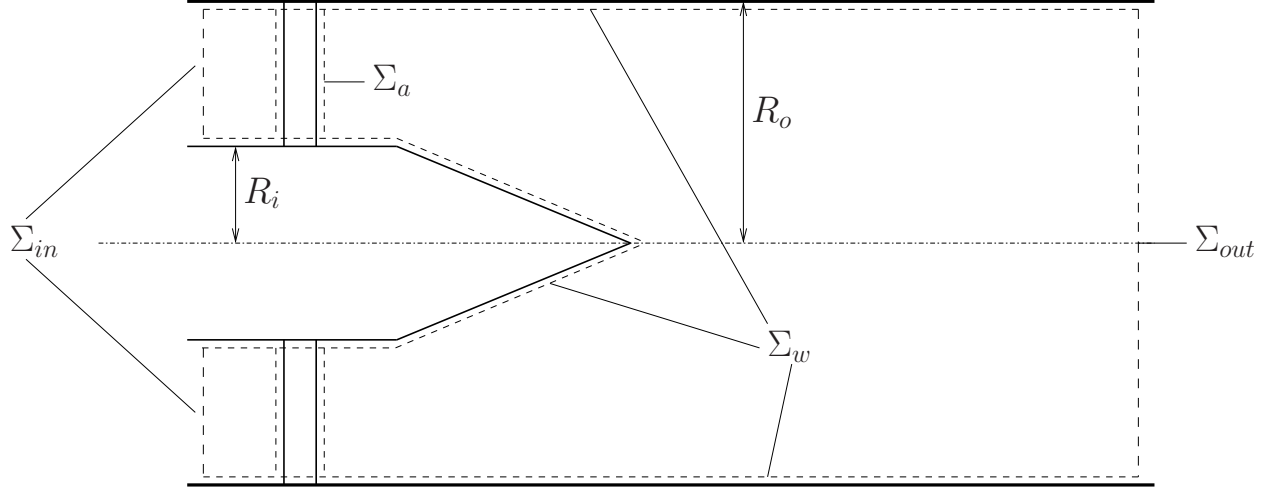
$$\frac{d}{dt} \left[\int_{V_c(t)} \rho \bar{v} dV \right] + \int_{\Sigma_c(t)} \rho \bar{v}(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = - \int_{\Sigma_c(t)} p \bar{n} d\sigma + \int_{\Sigma_c(t)} \bar{\tau}' \cdot \bar{n} d\sigma + \int_{V_c(t)} \rho \bar{f}_m dV$$

$$\begin{aligned} & \frac{d}{dt} \left[\int_{V_c(t)} \rho(\bar{x} - \bar{x}_o) \wedge \bar{v} dV \right] + \int_{\Sigma_c(t)} \rho[(\bar{x} - \bar{x}_o) \wedge \bar{v}][(\bar{v} - \bar{v}_c) \cdot \bar{n}] d\sigma = \\ & = - \int_{\Sigma_c(t)} p(\bar{x} - \bar{x}_o) \wedge \bar{n} d\sigma + \int_{\Sigma_c(t)} (\bar{x} - \bar{x}_o) \wedge (\bar{\tau}' \cdot \bar{n}) d\sigma + \int_{V_c(t)} \rho(\bar{x} - \bar{x}_o) \wedge \bar{f}_m dV \end{aligned}$$

$$\begin{aligned} & \frac{d}{dt} \left[\int_{V_c(t)} \rho(e + |\bar{v}|^2/2) dV \right] + \int_{\Sigma_c(t)} \rho(e + |\bar{v}|^2/2)(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = - \int_{\Sigma_c(t)} p \bar{v} \cdot \bar{n} d\sigma \\ & + \int_{\Sigma_c(t)} \bar{v} \cdot \bar{\tau}' \cdot \bar{n} d\sigma + \int_{V_c(t)} \rho \bar{f}_m \cdot \bar{v} dV - \int_{\Sigma_c(t)} \bar{q} \cdot \bar{n} d\sigma + \int_{V_c(t)} (Q_c + Q_r) dV \end{aligned}$$

Solution:

To solve the problem we make use of the mass, momentum, angular momentum and energy conservation equations in their integral form, applied to a fixed control volume, V_0 . Its associated control surface, $\Sigma_0 = \Sigma_{in} \cup \Sigma_{out} \cup \Sigma_w \cup \Sigma_a$, shown in the figure below, is decomposed into an annular inlet section Σ_{in} , a circular outlet section Σ_{out} , a surface containing the inner wall of the casing and the outer wall of the diffuser cone, Σ_w , and several surfaces surrounding the stator aerofoils, Σ_a .



The integral mass conservation equation,

$$\frac{d}{dt} \left[\int_{V_0} \rho dV \right] + \int_{\Sigma_0} \rho \bar{v} \cdot \bar{n} d\sigma = 0,$$

can be simplified by taking into account that the mass inside the control volume is a constant, so that the first term is zero, and that the only control surfaces crossed by the gas are Σ_{in} and Σ_{out} , thus obtaining

$$\int_{\Sigma_{in}} \rho \bar{v} \cdot \bar{n} d\sigma + \int_{\Sigma_{out}} \rho \bar{v} \cdot \bar{n} d\sigma = 0,$$

and, since the density and velocity are uniform throughout both surfaces,

$$\boxed{\rho_1 u_1 \pi (R_e^2 - R_i^2) = \rho_2 u_2 \pi R_e^2 = G} \quad (1)$$

where G is the mass flux, and we have taken into account that at the inlet $\bar{v} \cdot \bar{n} = -u_1$ and $\int_{\Sigma_{in}} d\sigma = \pi (R_e^2 - R_i^2)$.

Thanks to the axisymmetry of the problem, the only relevant component of the momentum equation is the axial component z , that writes

$$\frac{d}{dt} \left[\int_{V_0} \rho \bar{v} \cdot \bar{e}_z dV \right] + \int_{\Sigma_0} \rho (\bar{v} \cdot \bar{e}_z) (\bar{v} \cdot \bar{n}) d\sigma = - \int_{\Sigma_0} p \bar{n} \cdot \bar{e}_z d\sigma + \int_{\Sigma_0} \bar{e}_z \cdot \bar{\tau}' \cdot \bar{n} d\sigma,$$

where we have neglected the body forces. The first term vanishes for the same reasons discussed above and, again, the only non-zero contributions of the second term are the inlet and outlet convective momentum fluxes. In addition, viscous stresses are zero over the inlet and outlet sections, since the velocity field is locally uniform. The axial momentum equation then simplifies to

$$\int_{\Sigma_{in} \cup \Sigma_{out}} \rho (\bar{v} \cdot \bar{e}_z) (\bar{v} \cdot \bar{n}) d\sigma = - \int_{\Sigma_{in} \cup \Sigma_{out}} p \bar{n} \cdot \bar{e}_z d\sigma + \int_{\Sigma_w \cup \Sigma_a} \bar{e}_z \cdot (-p \bar{n} + \bar{\tau}' \cdot \bar{n}) d\sigma,$$

where the last integral represents the force that the inner solid walls exert on the gas, $F^{(s \rightarrow g)} = -F^{(g \rightarrow s)}$, where $F^{(g \rightarrow s)}$ is, by Newton's third law, the force that the gas exerts on the walls. The net force on the module, F , is obtained by adding the external force on the casing due to the atmosphere surrounding the module, $F = F^{(g \rightarrow s)} + F_{ext}$. In turn, it is easy to see that $F_{ext} = 0$ due to the axial symmetry. On the other hand, the integrals of the axial momentum flux at the inlet and outlet sections, as well as the corresponding axial pressure force, are all immediately calculated by taking into account the uniform conditions, giving the result

$$\boxed{F = \pi (R_e^2 - R_i^2) (p_1 + \rho_1 u_1^2) - \pi R_e^2 (p_2 + \rho_2 u_2^2)} \quad (2)$$

Taking the origin of momenta at any point along the axis, the integral angular momentum equation simplifies to a balance between the net angular momentum flux, and the torque of the pressure and viscous forces,

$$\int_{\Sigma_{in} \cup \Sigma_{out}} \rho (\bar{x} \wedge \bar{v}) (\bar{v} \cdot \bar{n}) d\sigma = - \int_{\Sigma_{in} \cup \Sigma_{out}} \bar{x} \wedge (p\bar{n}) d\sigma + \int_{\Sigma_w \cup \Sigma_a} \bar{x} \wedge (-p\bar{n} + \bar{\tau}' \cdot \bar{n}) d\sigma,$$

where we have taken into account the same simplifications discussed above for the mass and axial momentum equations, namely steady flow, fixed control volume, and uniform properties over the inlet and outlet sections. Due to the symmetry of the problem, it is easy to see that the outlet angular momentum flux, as well as the momenta of the pressure forces at both the inlet and outlet sections, are all zero, while the inlet angular momentum flux is associated to the azimuthal velocity component, $u_\theta \bar{e}_\theta$, and gives the contribution

$$\int_{\Sigma_{in}} \rho (\bar{x} \wedge u_\theta \bar{e}_\theta) (\bar{v} \cdot \bar{n}) d\sigma = \rho_1 u_1 u_\theta \bar{e}_z \int_{R_i}^{R_e} 2\pi r^2 dr = \frac{2}{3} \pi \rho_1 u_1 u_\theta (R_e^3 - R_i^3) \bar{e}_z,$$

where it has been taken into account that $\bar{x} \wedge \bar{e}_\theta = -r\bar{e}_z$ and $d\sigma = 2\pi r dr$. Moreover, the integral

$$\int_{\Sigma_w \cup \Sigma_a} \bar{x} \wedge (-p\bar{n} + \bar{\tau}' \cdot \bar{n}) d\sigma = -M,$$

where use has been made of Newton's third law, and where it has been taken into account that the external torque of the atmospheric forces on the casing is zero due to axisymmetry, $\bar{M}_{ext} = 0$. Therefore, the third equation is

$$\boxed{M = -\frac{2\pi}{3} \rho_1 u_1 u_\theta (R_e^3 - R_i^3)} \quad (3)$$

The only unknown in equation (3) is the torque on the diffuser module M , and can be obtained from the latter result alone.

The energy equation, simplified along the same lines as for the rest of the conservation equations above, takes the form

$$\int_{\Sigma_{in} \cup \Sigma_{out}} \rho (e + |\bar{v}|^2/2) \bar{v} \cdot \bar{n} d\sigma = - \int_{\Sigma_{in} \cup \Sigma_{out}} p \bar{v} \cdot \bar{n} d\sigma + \int_{\Sigma_w \cup \Sigma_a} \bar{v} \cdot (-p\bar{n} + \bar{\tau}' \cdot \bar{n}) d\sigma,$$

where the following facts have been used: there are no volumetric heat sources, $\bar{q} = -k\nabla T = 0$ at all the control surfaces (inlet and outlet due to uniform conditions, and casing and aerofoils due to the associated adiabatic conditions), and the work of the body forces has been neglected. The integral $\int_{\Sigma_w \cup \Sigma_a} \bar{v} \cdot (-p\bar{n} + \bar{\tau}' \cdot \bar{n}) d\sigma$ represents the power developed by the forces acting on the inner walls of the module, and is zero since they are at rest ($\bar{v} = 0$ over Σ_w and Σ_a). Finally, the first term in the right-hand-side, representing the power developed by the pressure forces at the inlet and outlet, may be combined with the energy flux on the left-hand-side, to give the net flux of stagnation enthalpy, $h + |\bar{v}|^2/2$, which is therefore a conserved magnitude,

$$\int_{\Sigma_{in} \cup \Sigma_{out}} \rho (h + |\bar{v}|^2/2) \bar{v} \cdot \bar{n} d\sigma = 0,$$

where $h = e + p/\rho = c_p T = \gamma/(\gamma - 1) p/\rho$ is the enthalpy per unit mass. Performing the integrals, one finally obtains,

$$\boxed{\frac{\gamma}{\gamma - 1} \frac{p_1}{\rho_1} + \frac{u_1^2 + u_\theta^2}{2} = \frac{\gamma}{\gamma - 1} \frac{p_2}{\rho_2} + \frac{u_2^2}{2}} \quad (4)$$

Equations (1), (2) and (4), together with the equation of state, $p = \rho R_g T$ determine the unknowns ($u_2, \rho_2, T_2, \bar{F}$), while $\bar{M}$ is directly given by equation (3).